

DAX MEASURES DOCUMENT

Securitisation Risk Assessment & Investor Reporting Dashboard

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Tools Used:

- Microsoft Power BI
- DAX (Data Analysis Expressions)
- Excel
- Data Modeling

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Introduction

This document contains the DAX measures developed for the Securitisation Risk Assessment and Investor Reporting Dashboard. The measures are used to calculate portfolio performance metrics, credit risk indicators, delinquency statistics, IFRS 9 Expected Credit Loss (ECL), collections, recoveries, vintage analysis, and investor reporting metrics. These calculations enable effective portfolio monitoring and data-driven decision making.

1. Executive Summary

This document contains the DAX measures developed for the Auto Loan Securitisation Risk Assessment Dashboard.

The measures support:

- Portfolio Monitoring
- Credit Risk Assessment
- Delinquency Tracking
- IFRS 9 Expected Credit Loss
- Collection & Recovery Monitoring
- Vintage Analysis
- Investor Reporting

More than 80 DAX measures were created to enable dynamic reporting and business decision-making.

2. Executive Dashboard Measures

Portfolio Outstanding

Portfolio Outstanding =

SUM(loan_master[CurrentBalance])

Purpose:

Measures total outstanding balance of all active loans.

Total Customers

Total Customers =

DISTINCTCOUNT(loan_master[CustomerID])

Purpose:

Calculates unique customer count.

Total Loans

Total Loans =

COUNTROWS(loan_master)

Purpose:

Displays total number of loans in portfolio.

Collection Efficiency %

Collection Efficiency % =

DIVIDE(
SUM(collections[ActualCollection]),
SUM(collections[ScheduledCollection])
)

Purpose:

Measures how efficiently scheduled collections were recovered.

Collection Status

Collection Status =

VAR CE = [Collection Efficiency %]

RETURN

SWITCH(
TRUE(),
CE > 0.95,"Excellent",
CE >=0.90,"Watch",
"Critical"
)

Purpose:

Traffic-light indicator for collection performance.

3. Credit Risk Dashboard Measures

Weighted Average CIBIL

Weighted Avg CIBIL =
DIVIDE(
SUMX(
loan_master,
loan_master[CIBILScore] *
loan_master[CurrentBalance]
)
[Portfolio Outstanding]
)

Purpose:
Portfolio-level weighted credit quality indicator.

Weighted Average LTV

Weighted Avg LTV =
DIVIDE(
SUMX(
loan_master,
loan_master[LTV] *
loan_master[CurrentBalance]
)
[Portfolio Outstanding]
)

Purpose:
Measures collateral coverage risk.

Weighted Average DTI

Weighted Avg DTI =
DIVIDE(
SUMX(
loan_master,
loan_master[DTI] *

```
loan_master[CurrentBalance]
```

```
),
```

```
[Portfolio Outstanding]
```

```
)
```

Purpose:

Measures borrower repayment burden.

4. Delinquency Dashboard Measures

Delinquent Loans

Delinquent Loans =

```
CALCULATE(
```

```
COUNTROWS(dpd_snapshot_history),
```

```
dpd_snapshot_history[CurrentDPD] > 0
```

```
)
```

Purpose:

Counts loans currently overdue.

Total Overdue Amount

Total Overdue Amount =

```
SUM(dpd_snapshot_history[AmountOverdue])
```

Purpose:

Measures total overdue exposure.

Cure Rate

Cure Rate =

```
DIVIDE(
```

```
COUNTROWS(
```

```
FILTER(
```

```
dpd_snapshot_history,
```

```
dpd_snapshot_history[RollFlag]="Better"
```

```
)
```

),
COUNTROWS(dpd_snapshot_history)
)

Purpose:

Measures percentage of loans improving delinquency status.

5. IFRS 9 Dashboard Measures

Stage 1 Exposure

Stage 1 Exposure =

CALCULATE(
[Portfolio Outstanding],
loan_master[IFRS9_Stage]=1
)

Stage 2 Exposure

Stage 2 Exposure =

CALCULATE(
[Portfolio Outstanding],
loan_master[IFRS9_Stage]=2
)

Stage 3 Exposure

Stage 3 Exposure =

CALCULATE(
[Portfolio Outstanding],
loan_master[IFRS9_Stage]=3
)

Total ECL

Total ECL =

SUM(ifs9_ecl[ECL_Provision])

Purpose:

Measures expected credit losses under IFRS 9.

6. Collections & Recovery Dashboard**Total Collections**

Total Collections =

SUM(collections[ActualCollection])

Recoveries

Recoveries =

SUM(dynamic_loss_monthly[Recoveries])

Recovery Rate

Recovery Rate =

DIVIDE(
[Recoveries],
SUM(dynamic_loss_monthly[GrossLoss])
)

Net Loss

Net Loss =

SUM(dynamic_loss_monthly[NetLoss])

7. Vintage Analysis Dashboard**Latest Pool Factor**

Latest Pool Factor =

MIN(static_pool_vintage_data[PoolFactor])

Purpose:

Measures remaining collateral balance relative to original pool balance.

Latest Net Loss Rate

Latest Net Loss Rate =

MAX(static_pool_vintage_data[CumulativeNetLossRate])

Latest Prepayments

Latest Prepayments =

MAX(static_pool_vintage_data[CumulativePrepayments])

8. Investor Reporting Dashboard

BOP Balance

BOP Balance =

SUM(pool_reporting[BOP_Balance])

EOP Balance

EOP Balance =

SUM(pool_reporting[EOP_Balance])

Gross Loss

Gross Loss =

SUM(pool_reporting[GrossLoss])

Net Loss Rate

Net Loss Rate =

DIVIDE(
[Net Loss],
[BOP Balance]
)

CPR

CPR =

AVERAGE(pool_reporting[CPR_Annualised])

SMM

SMM =

AVERAGE(pool_reporting[SMM])

HHI

HHI =

SUMX(
VALUES('auto_loan_securitisation_data'[State]),
POWER(
DIVIDE(
[Portfolio Outstanding],
CALCULATE(
[Portfolio Outstanding],
ALL('auto_loan_securitisation_data'[State])
)
),
2
)
)
)
* 10000